

# An instant private network that rescuers carry in

The phone a survivor already has becomes a two-way SOS beacon within minutes of a rescue team arriving. No towers, internet, app or SIM.

~₹1,750

per rescuer node incl.  
battery & case (est.)

~₹42k

20-node neighbourhood kit,  
hardware (est.)

-78%

airtime per SOS with AI  
compression

0

recurring telecom or cloud  
cost

## Problem

After earthquakes, floods and cyclones, power and cellular fail first. Survivors still have phones but no network. Government alerts are one-way, and existing mesh radios need a device per person set up in advance.

## Solution

A rescue kit: rescuer nodes (ESP32 + LoRa) that open a Wi-Fi SOS page for any phone and count nearby phones for a search heatmap; AI compression; quantum-planned, self-healing relays; and an offline command dashboard with Read/Dispatch status back to the survivor.

## Customer

Government responders: NDRF, 16 SDRFs, district disaster management authorities, fire services. Also large-event organisers, mines, coastal fisheries and defence units.

## Model

Kit sales through government procurement (GeM), an annual command-software licence, installation and training contracts, and event rentals.

All prices are estimates from typical Indian retail listings for single units (2026). Volume pricing is lower. Validate with supplier quotes before any bid.

# Hardware and software, layer by layer

**Survivor's phone** · any smartphone browser · captive-portal SOS page (HTML/JS, served by the node) · English / हिंदी / తెలుగు · tone + vibration on status

**Rescuer node** · ESP32-WROOM-32 (Wi-Fi AP + DNS captive portal + WebServer) · Ai-Thinker RA-02 SX1278 LoRa · Wi-Fi probe-request presence counter (promiscuous mode, hashed, 20 s window) · retry/ACK queue · LEDs + push button · Arduino C++

**Relays** · same board in relay role · HELLO beacons, hop gradient, closer-only forwarding, retry-next-best, store-and-carry · positions planned by QAOA

**Command gateway** · ESP32 + RA-02 · LoRa ⇄ USB JSON lines · ACK / READ / DISPATCH / REPLY back over LoRa

**Command laptop** · Python 3 standard-library server · SQLite log · Leaflet map with bundled GHMC ward boundaries · Web Audio + speech alerts · viewer role over local Wi-Fi · CAP 1.2 export (proposed)

**Intelligence** · SOS AI: multilingual Naive Bayes ensemble (17 heads), SemCode 40-bit codebook, Semtech airtime model · Quantum: XY-mixer QAOA in Qiskit 2.5 + Aer, COBYLA, brute-force verification · Self-healing simulator

| SOFTWARE       | TECH                                                                       | WHY                                                          |
|----------------|----------------------------------------------------------------------------|--------------------------------------------------------------|
| Firmware       | Arduino C++, ESP32 core 3.x, LoRa (Sandeep Mistry), WebServer, DNSServer   | Standard, well-tested, flashes with arduino-cli              |
| Radio settings | 433 MHz (865 MHz in production), SF9, 125 kHz, CR 4/5, CRC, sync word 0x5A | ~1.8 kbit/s; good sensitivity; ignores other LoRa traffic    |
| Command        | Python 3 (no installs), pyserial, SQLite, HTML/JS, Leaflet                 | Runs offline on any laptop; phones open it over local Wi-Fi  |
| AI             | Pure-Python Naive Bayes, multilingual lexicon features                     | Tiny, offline, deterministic, retrains in 3 s                |
| Quantum        | Qiskit 2.5, Qiskit Aer, SciPy                                              | Industry-standard; same code can target IBM quantum hardware |

COST

Bill of materials (estimated, Indian retail, single units)

| ITEM                                                         | USED IN              | UNIT ₹  |
|--------------------------------------------------------------|----------------------|---------|
| ESP32-WROOM-32 dev board                                     | every node           | 450     |
| Ai-Thinker RA-02 LoRa SX1278 (433 MHz)                       | every node           | 500     |
| 433 MHz antenna + pigtail                                    | every node           | 150     |
| PCB / wiring, LEDs, push button                              | every node           | 100     |
| Electronics per node                                         |                      | ≈ 1,200 |
| 10,000 mAh power bank (≈ 2–3 days with hotspot on, estimate) | rescuer node / relay | 400     |
| IP65 enclosure                                               | field nodes          | 150     |
| Deployable node                                              |                      | ≈ 1,750 |

Kit sizes

| KIT               | CONTENTS                               | HARDWARE ₹ |
|-------------------|----------------------------------------|------------|
| Demo kit (today)  | 2 ESP32, 2 RA-02, antennas, power bank | ≈ 2,800    |
| Team kit          | 4 rescuer nodes, 6 relays, 1 gateway   | ≈ 19,300   |
| Neighbourhood kit | 20 nodes + 2 gateways, rugged cases    | ≈ 42,000   |

The laptop is the agency's existing one; the software has no licence cost for the prototype.

Compared with today's options

Satellite handset (one responder)  

≈ ₹1,34,000 (BSNL Inmarsat handset, reported)

VOID-NAV neighbourhood kit (20 nodes)  

≈ ₹42,000 • covers any number of survivors' phones

One VOID-NAV node  

≈ ₹1,750

# Who pays, for what

| STREAM                          | CUSTOMER                                                   | MODEL                                                                |
|---------------------------------|------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Rescue kits</b>              | NDRF, SDRFs, district disaster authorities, fire services  | One-time hardware sale via government procurement (GeM)              |
| <b>Command software licence</b> | State disaster management authorities                      | Annual licence: dashboard, AI model updates, relay planner, training |
| <b>Pre-placed SOS points</b>    | Cyclone shelters, schools, hospitals, panchayats, railways | Installation + maintenance contracts (government / CSR funded)       |
| <b>Event mode</b>               | Kumbh-scale gatherings, stadiums, pilgrimages              | Per-event rental: lost-person and medical SOS when networks jam      |
| <b>Defence variant</b>          | Army / paramilitary units without infrastructure           | Through iDEX and defence-innovation programmes                       |

## Why it's viable

Cheap nodes; licence-free radio band; no SIMs or cloud; uses phones people already own. No device distribution to citizens.

## Fits India's system

Exports standard CAP 1.2 records, the format of NDMA's SACHET platform, so it adds to existing control rooms instead of replacing them. (Proposed.)

## Go-to-market

Pilot drill with one district authority or SDRF → case study → state rate contract → GeM listing → CSR-funded shelter installs.

## Competition

| CAPABILITY                             | SATELLITE SOS | MESHTASTIC | PROJECT OWL | VOID-NAV         |
|----------------------------------------|---------------|------------|-------------|------------------|
| Uses survivor's existing phone, no app | Some phones   | ✗          | ✓           | ✓                |
| Survivor sees Read + Dispatched        | Partial       | ✗          | ✗           | ✓                |
| Local rescue dispatch workflow         | Remote centre | ✗          | Cloud       | ✓ <b>offline</b> |
| Bandwidth-aware AI compression         | —             | ✗          | ✗           | ✓                |
| Relay planning + self-healing          | —             | Partial    | Partial     | ✓                |

# From hackathon kit to standard rescue equipment

|            |                                                                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Now        | Working survivor ↔ rescuer loop · rescuer node, gateway and USB-link firmware · offline dashboard · trained multilingual SOS model · QAOA planner in Qiskit · self-healing simulator.        |
| +1 month   | Third radio for a physical multi-hop relay · field range test · GPS module per node · SemCode on air in firmware · AES-128 + authentication tag on every packet · flash-saved message queue. |
| +3 months  | 865–867 MHz modules (WPC licence-free band) · waterproof enclosures · solar relays · cached offline map tiles per district · retrain the AI on real drill messages.                          |
| +6 months  | Pilot drill with a district authority or SDRF · multi-gateway database sync · live CAP export to SACHET / state control room / 108 over satellite backhaul.                                  |
| +12 months | Drone or tethered-balloon relays for wide areas · QAOA on IBM quantum hardware for city-scale relay plans · privacy-preserving survivor-presence radar · rescuer phone app.                  |

## Risks and how we handle them

| RISK                                                        | MITIGATION                                                                        |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Phones with Wi-Fi off / dead battery can't be found or send | Push-button SOS on the node; works alongside dogs, radar and thermal search       |
| Spoofed SOS / eavesdropping                                 | Per-deployment keys + packet authentication (roadmap); no personal data collected |
| Radio regulation                                            | Move to 865–867 MHz with duty-cycle limits; one-line firmware change              |
| AI misreads a message                                       | Taps always win; AI can only raise severity; survivor's own words always shown    |
| Procurement cycles are slow                                 | Start with CSR-funded shelter pilots and event rentals for early revenue          |